

Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

SESSION: 2025-26



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Abstract

The Automated Teller Machine (ATM) Simulation is a console-based Java application developed using Object-Oriented Programming (OOP) concepts. The project simulates the basic functionality of an ATM system where users can securely access their account using a PIN. After successful authentication, users can perform operations such as withdrawing money and checking their account balance.

The project is designed to demonstrate the practical implementation of OOP concepts like classes, objects, encapsulation, and exception handling. It also provides secure handling of account information and validates user inputs to prevent errors. Exception handling is used to manage situations such as incorrect input, insufficient balance, and invalid withdrawal amounts.

This project helps students understand real-world banking systems and how Java can be used to create secure and interactive applications. The system is simple, user-friendly, and can be enhanced in the future by adding features such as deposit functionality, multiple user accounts, transaction history, and database connectivity.

Introduction

Automated Teller Machines (ATMs) are electronic banking systems that allow customers to perform banking transactions without visiting a bank branch. This project develops a simple ATM simulation using Java programming language and Object-Oriented Programming principles.

The system authenticates users through a PIN verification process and provides options such as money withdrawal and balance inquiry. The project demonstrates secure data management using encapsulation and handles runtime errors through exception handling.

Features

- PIN Authentication
- Money Withdrawal
- Balance Inquiry
- Secure Account Data Handling
- Exception Handling
- Menu Driven Interface

Objectives

- To simulate the working of an ATM system.
- To implement Object-Oriented Programming concepts.

- To provide secure PIN authentication.
- To manage account balance securely.
- To perform withdrawal transactions.
- To display account balance.
- To handle invalid user inputs using exception handling.
- To develop a menu-driven Java application.

Scope

This project is suitable for educational purposes and small-scale banking simulations. It helps students understand the implementation of OOP concepts in real-world applications. The project can be further extended with database integration, multiple user accounts, fund transfers, and transaction history management.

System Requirements

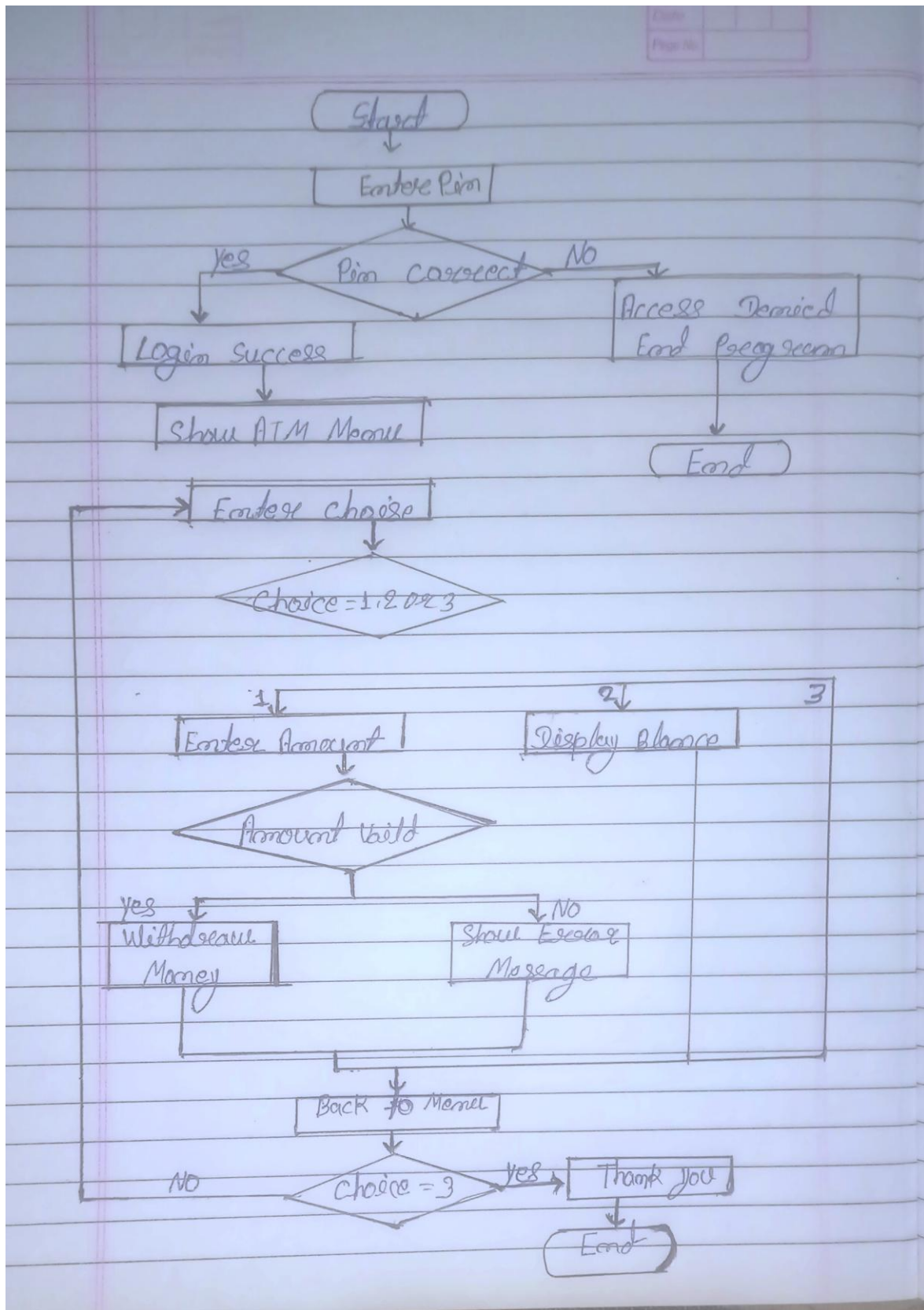
Hardware: Computer with 4GB RAM

Software: Java JDK, IDE (VS Code/Eclipse/Edit Plus), Windows

Methodology

1. User enters ATM PIN.
2. System verifies the PIN.
3. If PIN is incorrect, access is denied.
4. If PIN is correct, ATM menu is displayed.
5. User selects an option:
 - Withdraw Money
 - Check Balance
 - Exit
6. For withdrawal:
 - Amount is validated.
 - Balance availability is checked.
 - Amount is deducted from account balance.
7. Balance inquiry displays current account balance.
8. Exception handling manages invalid inputs and errors.
9. User exits the system.

Flowchart



ClassName

Class: Account

Variables:

- balance

Methods:

- Account()
- withdraw()
- getBalance()

Array:

- No array used
-

Class: ATM

Variables:

- PIN

Methods:

- authenticate()
- showMenu()

Array:

- No array used
-

Class: ATMSimulation

Variables:

- Scanner sc
- Account account
- ATM atm
- choice
- enteredPin

Methods:

- main()

Array:

- String[] args

Implementation

Account Class

The Account class manages the user's account balance. The balance variable is declared private to maintain security. It provides methods to withdraw money and retrieve the current balance. The withdrawal method validates the amount and

checks for sufficient balance before processing the transaction.

ATM Class

The ATM class manages user authentication and menu display. It contains a static PIN used for verification. The authenticate() method checks whether the entered PIN is correct, while the showMenu() method displays available ATM operations.

ATMSimulation Class

This class contains the main method. It creates objects of ATM and Account classes, takes user input, validates PIN, displays menu options, and performs operations according to the user's choice. Exception handling is used to manage invalid inputs and prevent program crashes.

Sample Input / Output

Output 1

Enter PIN: 1234

ATM MENU

1. Withdraw Money
2. Check Balance
3. Exit

Enter your choice: 1

Enter amount to withdraw: 500

Transaction successful.

Output 2

Enter your choice: 2

Current Balance: ₹1500

Output 3

Enter amount to withdraw: 5000

Error: Insufficient balance.

Output 4

Enter your choice: 3

Thank you for using ATM.

Future Enhancements

- Deposit Money Feature
- Multiple User Accounts

- Transaction History
- Mini Statement Generation
- Database Connectivity
- Fund Transfer Facility
- OTP Based Authentication
- Graphical User Interface (GUI)

```
> cd "c:\U
Enter PIN: 1234
Login Successful!

ATM MENU
1. Withdraw Money
2. Check Balance
3. Exit
Enter your choice: 2
Current Balance: ?2000.0

ATM MENU
1. Withdraw Money
2. Check Balance
3. Exit
Enter your choice: 1
Enter amount to withdraw: 200
Transaction successful.

ATM MENU
1. Withdraw Money
2. Check Balance
3. Exit
Enter your choice: 1
Enter amount to withdraw: [ ]
```

Gantt Chart

1. Planning & Structure

The project was divided into different phases such as planning, system design, coding, testing, documentation, and final submission. This helped in organizing the work properly and completing the project systematically.

2. Time Management

The Gantt Chart shows the starting and ending time of each task. It helps in tracking progress, avoiding delays, and completing the project on time.

Task	Week 1	Week 2	Week 3	Week 4
Project Planning	✓			
System Design	✓	✓		
Coding & Development		✓	✓	
Testing & Debugging			✓	
Documentation				✓
Final Submission				

Conclusion

The Automated Teller Machine Simulation project successfully demonstrates the application of Object-Oriented Programming concepts in Java. The project provides secure PIN authentication, balance management, and withdrawal operations while utilizing encapsulation and exception handling. It offers a

practical understanding of real-world banking systems and serves as a strong foundation for developing more advanced financial applications.

References

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